EE-0129 Project 2 Report
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Introduction

This project submissions includes a python program, “EE129 Project 2 Source Code - Michael Zhang.py”, the program’s associated .html and .txt files, and a separate folder containing modified demo code. The main project source code creates a server with a default page containing a color picker, and a secondary “login” page as outlined by the project requirements. The project requirements, such as how the server handles the color picker pages and login handling pages, and manual user interrupts are hardcoded into the program. All other server functionality like http responses is dynamically handled using the “mimetypes” library and the receiveMessage function, which will be further discussed in the Extra Credit section. The supporting .html webpages use this extra functionality extensively to add a bit more flair to the project requirements.

The modified demo code creates a barebones server that displays whatever content was requested in the URL that can be handled by http.

Basic Program Function

To run “EE129 Project 2 Source Code - Michael Zhang.py”, extract the full source code from the zip attached to this project’s submission, or from github:
https://github.com/EngineerIsGaming1/Project-2

Afterwards navigate to the directory location containing the source code in the windows command prompt and enter “python (or py, or python3 depending on your python install) EE129 Project 2 Source Code - Michael Zhang.py”.

When the source code is run, the terminal will display the message “The server is ready to receive”. Once this message has been displayed, the python server is active and ready for use. Using a web browser, navigate to localhost:6789 to interact with the server. The default page when no extensions are added to the URL is the color picker. Adding “/login.html” will take you to the login page, and all other URLs will result in a 404 not found error.

Inside the windows terminal, “close” can be inputted to close the server and exit the program. Ctrl+C will also terminate the server and program. “help” will give basic instructions on what “close” does, and all other inputs result in an error message.

Results and Discussion

This project was extremely interesting an engaging. It was satisfying to apply all the concepts learned in class, and experience firsthand how well these standard protocols like http work. For example, the demo code made it seem like servers required a lot of hardcoded case statements to function. However, after reviewing project slides, textbook content, and reviewing all the functions used in the demo code using the python documents and w3Schools tutorials, I was able to create a demo code that only needed a few lines of code to respond to any type of http content request through the use of the mimetypes library.

Extra Credit

The extra functionality implemented in the main project source code is the ability to handle multiple content requests outside of what is required by the project. Files like login_failed.html and login_success.html make additional http GET requests for .gif files in addition to the original content request, and the program is able to handle the request without issue.